Curvature Identities and Restricted Potential Bounds in Einstein–Cartan–Nieh–Yan Minisuperspace with Irreducible EC-Weyl Corrections

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Abstract

We evaluate the Einstein–Cartan (EC) action with a constant complete Nieh–Yan (NY) coupling and an irreducible EC-Weyl-squared term on static homogeneous families of $S^3 \times S^1$, $T^3 \times S^1$, and a fixed compact $\mathrm{Nil}^3 \times S^1$. Varying radius and volume-preserving shape at fixed torsion parameters and circle circumference, we classify boundedness and attainment of minima of the reduced potential. The complete NY density and the form-index Pontryagin density vanish for the prescribed connection, while the Weyl contribution is governed by the spatial shape dependence. We prove that the irreducible EC-Weyl tensor equals the Levi-Civita (LC) Weyl tensor throughout these families, so their reduced potentials and classifications coincide. A brief comparison isolates the additional curvature sectors in a different Ricci-tracefree quadratic action. For S^3 at nonpositive Weyl coupling, an exact factorization of the shape difference reduces global minimization in two variables to the round slice. A unique finite-radius minimum exists for $V \neq 0$ and $\eta^2 < 4$; its radius and value are independent of the Weyl coupling. The remaining parameter regions distinguish unattained infima, a degenerate minimum set, and a large-radius runaway. Positive coupling produces small-radius unboundedness for nonround spheres and Nil. Comparison with the shape-independent torus potential and the monotone Nil shape dependence identifies how geometry and variation paths determine the extrema. These results concern the specified static homogeneous reductions with the stated fixed data.

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1 Introduction

1.1 Problem and motivation

Understanding a curvature-squared contribution to a homogeneous reduced potential requires separating changes of overall radius from volume-preserving changes of shape. Vanishing of the Weyl term on a round slice and minimization with respect to deformations away from that slice are distinct questions. We study their relation for the Einstein–Cartan (EC) action with the complete Nieh–Yan (NY) term and an irreducible EC-Weyl-squared contribution on closed product manifolds.

A preceding study classifies radius potentials for EC plus complete NY at fixed torsion parameters and circle circumference, distinguishing geometries and variation paths.[1] Here we add a volume-preserving shape variable and compare Levi-Civita (LC) and irreducible EC-Weyl corrections to that classification. The spatial factors are S^3, T^3, and a fixed compact Heisenberg quotient; local connections and global integration domains are specified separately.

1.2 Main results

The curvature of the prescribed EC connection consists of spatial two-forms, giving a vanishing form-index Pontryagin density. The complete NY density also vanishes. Thus the reduced potential is determined by the EC curvature and torsion terms together with the Weyl term. We prove a tensor identity equating irreducible EC-Weyl and LC-Weyl curvature on the entire prescribed family. The actions built from their squared norms therefore yield the same reduced potential and classification of minima on this family, without identifying the unrestricted theories. We also compare an action built from the squared norm of Ricci-tracefree EC curvature, isolating its additional curvature contributions.

The central result is an exact decomposition of the spherical potential into its round value at the same radius and a nonnegative shape difference. For nonpositive Weyl coupling, equality in the shape bound holds only on the round slice. Global minimization in radius and shape reduces to a cubic polynomial in one variable. This gives the existence conditions, location, value, and restricted Hessian of the minimum analytically. The Weyl term preserves its round radius and value while increasing the curvature of the potential in the shape direction.

For positive coupling the inverse-radius term gives small-radius unboundedness on nonround spheres and Nil. In contrast, the Weyl term vanishes on the torus, while monotonicity in shape excludes a finite-shape minimum for Nil at nonpositive coupling. We compare these results under explicit local and global conventions, distinguishing attained minima from infima approached in degenerate limits.

1.3 Organization and scope

Section 2 defines the action, geometry, and fixed data. Sections 3–4 derive the curvature identities and reduced potentials. Sections 5–7 establish the spherical shape bound, positive-coupling asymptotics, and parameter classification; Section 8 compares the torus and Nil families. Section 9 relates the variational scope to physical questions, and Section 10 gives the conclusions. The appendices provide connection details, reproduction procedures, proof details, and the relation to the preceding version.

The minimization is restricted to static homogeneous families with positive radius and shape. The fixed quantities include signed torsion parameters and circle circumference. Full-field vacua and dynamical stability are distinct questions, discussed in Section 9.

2 Action, geometry, and fixed data

We use an orthonormal Euclidean metric δ_{ab} and orientation $\epsilon^{0123} = +1$. The constants κ, L are positive; α, θ are real. No cosmological term, matter source, or boundary is added. The main action contains the squared norm I_{W} of the irreducible EC-Weyl tensor C_{EC}, whose projection is defined in Section 4.2. We call this the irreducible EC-Weyl action. The LC-Weyl action replaces only this quadratic term by the squared norm $I_{\mathrm{LC}} = (C_{\mathrm{LC}})_{abcd}(C_{\mathrm{LC}})^{abcd}$; the EC Ricci scalar and complete NY form remain common. Section 4.4 compares the Ricci-tracefree EC curvature action. These names specify the quadratic term, not a change of the entire connection.

$$S = \int_M \left[\left(\frac{R_{\mathrm{EC}}}{2\kappa^2} + \alpha I_{\mathrm{W}} \right) \mathrm{vol}_4 + \theta N \right], \qquad U = -S\big|_{\mathrm{ansatz}}. \tag{A}$$

$I_{\mathrm{W}} = (C_{\mathrm{EC}})_{abcd}(C_{\mathrm{EC}})^{abcd}$. Selecting this curvature sector is a model choice, not a claim that other sectors are physically forbidden. The sign convention for U is explicit: it is minus the restricted action, not a Lorentzian Hamiltonian. Our local conventions are

$$\begin{aligned} de^a &= \frac{1}{2} C^a{}_{bc} e^b \wedge e^c, & \omega^a{}_b &= \Gamma^a{}_{bc} e^c, \\ T^a &= de^a + \omega^a{}_b \wedge e^b, & R^a{}_b &= d\omega^a{}_b + \omega^a{}_c \wedge \omega^c{}_b, \\ T^a &= \frac{1}{2} T^a{}_{bc} e^b \wedge e^c, & R_{ab} &= \frac{1}{2} R_{abcd} e^c \wedge e^d. \end{aligned} \tag{C}$$

The curvature–coframe term contains one two-form and two one-forms, giving coefficient $1/2$; the product of two torsion two-forms gives $1/4$.[2, 3]

$$\begin{aligned} N &= d(e^a \wedge T_a) = T^a \wedge T_a - R_{ab} \wedge e^a \wedge e^b, \\ \frac{N}{\mathrm{vol}_4} &= \frac{1}{4} \epsilon^{abcd} T^e{}_{ab} T_{ecd} - \frac{1}{2} \epsilon^{abcd} R_{abcd}. \end{aligned} \tag{NY}$$

Define the scalar coefficients of the two four-forms by

$$\begin{aligned} \mathcal{T}\,\mathrm{vol}_4 &= T^a \wedge T_a, \\ \mathcal{R}\,\mathrm{vol}_4 &= R_{ab} \wedge e^a \wedge e^b, \\ N &= (\mathcal{T} - \mathcal{R})\mathrm{vol}_4. \end{aligned} \tag{NYD}$$

Here $\mathcal{R}$ is the curvature–coframe contraction, not the Ricci scalar. The code selectors are related to these coefficients in Appendix B. Unless a comparison is stated explicitly, U denotes the reduced potential of the irreducible EC-Weyl action.

Introduce the volume-preserving coframe below. We use r for the scale of all three manifolds; it is the parameter called R by the Nil engine. The round sphere has $x = 1$.

$$\begin{aligned} e^0 &= r s^{1/3} \sigma^0, & e^1 &= r s^{1/3} \sigma^1, \\ e^2 &= r s^{-2/3} \sigma^2, & e^3 &= L\, du, \\ s &= 1 + \varepsilon > 0, & x &= s^{-2/3} > 0. \end{aligned} \tag{F}$$

$$
\begin{aligned}
de^0 &= n_0 e^1 \wedge e^2, & de^1 &= n_1 e^2 \wedge e^0, \\
de^2 &= n_2 e^0 \wedge e^1, & de^3 &= 0, \\
(n_0, n_1, n_2)_{S^3} &= \frac{1}{r}(4/x, 4/x, 4x^2), & & \\
(n_0, n_1, n_2)_{T^3} &= (0,0,0), \qquad (n_0, n_1, n_2)_{\mathrm{Nil}^3} = \frac{1}{r}(0,0,-x^2). & &
\end{aligned}
\tag{G}
$$

For S^3 we use SU(2), not a further quotient; its round physical radius is $r/2$. The spatial angles of T^3 have period 2π. The circle coordinate u has period one, so L is its circumference. For Nil use $\sigma^0 = dy^0$, $\sigma^1 = dy^1$, $\sigma^2 = dz - y^0 dy^1$, with periods $(2\pi, 2\pi, 4\pi^2)$ and the y^0-face gluing $z \mapsto z + 2\pi y^1$. Hold this quotient fixed when varying the scale and shape.

$$
\mathrm{Vol}_4 = c_M L r^3, \qquad c_{S^3} = \frac{\pi^2}{4}, \quad c_{T^3} = (2\pi)^3, \quad c_{\mathrm{Nil}^3} = (2\pi)^4. \tag{VOL}
$$

The geometric torsion is prescribed by

$$
T_{abc} = \frac{2\eta}{r}\epsilon_{abc}\big|_{a,b,c<3} + \frac{1}{3}(\delta_{ac}V_b - \delta_{ab}V_c), \qquad V_b = V\delta_{b3}. \tag{T}
$$

The spatial epsilon term is present only when all its indices are below three. We fix signed η, V and vary r and shape: the axial frame coefficient $2\eta/r$ is not held fixed as r varies. It is the same in all three spatial frame directions and does not depend on shape; this is part of the definition of the family. Referring it instead to one shape-dependent length, as the preceding study does on its anisotropic torus paths,[1] defines a different family that agrees with (T) only at $x = 1$. The public engine retains its signed input array τ; the dictionary is

$$
\begin{aligned}
T_{\mathrm{geom}} &= -\tau_{\mathrm{code}}, & (\eta, V) &= (-\eta_\tau, -V_\tau), \\
K_{abc}[T] &= -\frac{1}{2}(T_{abc} + T_{bca} - T_{cab}), & K_{abc}[\tau] &= +\frac{1}{2}(\tau_{abc} + \tau_{bca} - \tau_{cab}).
\end{aligned}
\tag{D}
$$

3 Complete NY and curvature two-forms

For (G), the spatial volume form is closed. Direct substitution of (T) gives (Q), using the scalar coefficients defined in (NYD). Simultaneous reversal of the two signed amplitudes leaves their product unchanged.

$$
e^a \wedge T_a = \frac{6\eta}{r} e^0 \wedge e^1 \wedge e^2, \qquad \mathcal{T} = \mathcal{R} = -\frac{4V\eta}{r}, \qquad N = 0. \tag{Q}
$$

Thus the complete NY density vanishes on this family and U is independent of θ. This does not assert a vanishing local density for arbitrary position-dependent torsion, actual boundaries, defects, or variable coupling. Using $\mathcal{T}$ or $\mathcal{R}$ alone gives distinct actions for component comparisons, rather than the complete NY action.

Proposition 1 (curvature forms of the prescribed connection). The EC connection one-forms for (G),(T) contain only spatial one-forms. Their exterior derivatives and products are spatial two-forms. Hence the circle-form components vanish and $R_{ab} \wedge R_{ab} = 0$.

To prove this, the LC and axial contortion one-forms are spatial, while the vector contortion contributes ω_{i3} proportional to e^i. The coefficients are constant in the invariant frame. This is a statement about this connection, not a consequence of a Killing direction alone.

$$\begin{aligned} \mathcal{P}\,\mathrm{vol}_4 &= \sum_{a,b} R_{ab}\wedge R_{ab}, \qquad \mathcal{P} = \frac{1}{4}\epsilon^{cdef}R_{abcd}R_{abef}, \\ E_{\mathrm{all}} &= \sum_{a,b,c,d} R_{abcd}^2, \qquad P_{\mathrm{all}} = \sum_{a,b,c,d} R_{abcd}(*R)_{abcd} = 2\mathcal{P}, \\ (*R)_{abcd} &= \frac{1}{2}\epsilon_{cdef}R_{abef}. \end{aligned} \tag{P}$$

Equation (P) uses all-component sums. E_{all} is a curvature norm, not the Euler density. The form-index Hodge star exchanges spatial and mixed two-forms. Consequently a nonzero curvature in this family is neither self-dual nor anti-self-dual; its two projected norms are equal.

The internal-index dual differs from the form-index dual. The internal contraction $\frac{1}{4}\epsilon^{abef}R_{abcd}R_{efcd}$ dualizes the first index pair instead of the form indices. At the round point $r = 3$, with code inputs $V_\tau = \eta_\tau = 1$, it equals $-16/27$, whereas $\mathcal{P} = 0$. Pair-exchange symmetry is not generally available for EC curvature, so (P) uses the form-index dual defined by exterior products. The geometric content of Proposition 1 is that the curvature of the prescribed connection belongs to the spatial two-form subspace.

4 Weyl projections, curvature identities, and reduced potentials

4.1 LC curvature as a reference

Solving the torsion-free Cartan equations with metric compatibility yields the LC connection and the following scalar for arbitrary n_i in (G). Changing from input-array signs to geometric torsion signs does not change these quadratic scalar terms.

$$\begin{aligned} R_{\mathrm{LC}} &= n_0n_1 + n_1n_2 + n_2n_0 - \frac{1}{2}(n_0^2 + n_1^2 + n_2^2), \\ R_{\mathrm{EC}} &= R_{\mathrm{LC}} - \frac{6\eta^2}{r^2} - \frac{2}{3}V^2. \end{aligned} \tag{R}$$

Using the four-dimensional LC identity $C^2 = R_{abcd}R^{abcd} - 2R_{ab}R^{ab} + R^2/3$ gives

$$\begin{aligned} f(x) &= 4x - x^4, \\ R_{\mathrm{LC},S^3} &= \frac{8f(x)}{r^2}, \qquad C^2_{\mathrm{LC},S^3} = \frac{1024x^2(1-x^3)^2}{3r^4}, \\ R_{\mathrm{LC},T^3} &= 0, \qquad C^2_{\mathrm{LC},T^3} = 0, \\ R_{\mathrm{LC},\mathrm{Nil}^3} &= -\frac{x^4}{2r^2}, \qquad C^2_{\mathrm{LC},\mathrm{Nil}^3} = \frac{4x^8}{3r^4}. \end{aligned} \tag{W}$$

4.2 Irreducible EC-Weyl curvature and the LC identity

For a metric-compatible connection, curvature is antisymmetric in each index pair, but pair-exchange symmetry and the torsion-free algebraic Bianchi identity need not hold. We select the Weyl part of the standard Riemann–Cartan decomposition.[4] In four Euclidean dimensions this

part has ten components, splitting into two five-dimensional Hodge sectors. Define the following linear operations on the full EC curvature:

$$\begin{aligned} S_{abcd} &= \tfrac{1}{2}(R^{\mathrm{EC}}_{abcd} + R^{\mathrm{EC}}_{cdab}), \\ F_{abcd} &= \tfrac{1}{3}(S_{abcd} + S_{acdb} + S_{adbc}), \qquad Q = S - F, \\ q_{bd} &= \delta^{ac} Q_{abcd}, \qquad s = \delta^{bd} q_{bd}, \\ (C_{\mathrm{EC}})_{abcd} &= Q_{abcd} - \tfrac{1}{2}(\delta_{ac} q_{bd} - \delta_{ad} q_{bc} - \delta_{bc} q_{ad} + \delta_{bd} q_{ac}) + \tfrac{s}{6}(\delta_{ac}\delta_{bd} - \delta_{ad}\delta_{bc}). \end{aligned} \tag{WP}$$

Here F is a four-form, and Q has algebraic Riemann symmetries. The last line is the unique orthogonal removal of the part reconstructed from its Ricci tensor using the metric: contraction gives $q_{bd} - (2q_{bd} + s\delta_{bd})/2 + s\delta_{bd}/2 = 0$. This defines a curvature sector without imposing torsion-free field equations.

Proposition 2 (identity on the prescribed family). For all $r, x > 0$ and all signed η, V in (T),

$$C_{\mathrm{EC}} = C_{\mathrm{LC}}, \qquad I_{\mathrm{W}} = I_{\mathrm{LC}}. \tag{ID}$$

Proof. Set $h = \eta/r$ and $q_0 = h^2 + V^2/9$. Using the Cartan connection and contortion in Appendix A, for arbitrary n_0, n_1, n_2 in (G) the independent components of $\Delta R = R_{\mathrm{EC}} - R_{\mathrm{LC}}$ are

$$\begin{aligned} \Delta R_{ijkl} &= -q_0(\delta_{ik}\delta_{jl} - \delta_{il}\delta_{jk}), \\ \Delta R_{i3jk} &= -\tfrac{2hV}{3}\epsilon_{ijk}, \\ \Delta R_{ijk3} &= 0, \qquad \Delta R_{i3j3} = 0, \qquad i,j,k,l \in \{0,1,2\}. \end{aligned} \tag{CD}$$

Other components follow from antisymmetry within each pair. The first line is entirely Ricci/scalar curvature: its spatial Ricci tensor is $-2q_0\delta_{ij}$, its 33 component is zero, and its scalar is $-6q_0$; substitution in the last line of (WP) gives zero. The pair-symmetric part of the second line is the four-form $-hV\epsilon_{abcd}/3$, removed in F. Its pair-antisymmetric part is removed in S. Thus the Weyl projection of ΔR vanishes. The LC curvature is unchanged by the first two operations in (WP), proving the tensor identity. No division by h, V or $x-1$ is used. Since (CD) is independent of the structure constants, the proof covers every positive shape in all three geometries. □

The identity concerns this static homogeneous family. It does not assert equality of the LC-Weyl and irreducible EC-Weyl actions for general fields, of their unrestricted variations, or of their time-dependent perturbation operators.

4.3 Common reduced potentials for the two Weyl actions

Substituting the volume and the vanishing complete NY density gives

$$U_{S^3} = \frac{c_{S^3} L}{3\kappa^2}\left[V^2 r^3 + (9\eta^2 - 12 f(x))r - \frac{1024\kappa^2\alpha x^2(1-x^3)^2}{r}\right]. \tag{S3}$$

$$\begin{aligned} U_{T^3} &= \frac{c_{T^3} L}{3\kappa^2}(V^2 r^3 + 9\eta^2 r), \\ U_{\mathrm{Nil}^3} &= \frac{c_{\mathrm{Nil}^3} L}{3\kappa^2}\left[V^2 r^3 + \left(9\eta^2 + \frac{3}{4}x^4\right) r - \frac{4\kappa^2\alpha x^8}{r}\right]. \end{aligned} \tag{TN}$$

Near the round slice, the Weyl term starts at quadratic order in the shape deformation. Distinguishing the quadratic coefficient from the second derivative gives the following local expansion.

$$\begin{aligned} C^2_{\mathrm{LC},S^3} &= \frac{1024\varepsilon^2(\varepsilon+2)^2}{3r^4(1+\varepsilon)^{16/3}} = \frac{4096}{3r^4}\varepsilon^2 + O(\varepsilon^3), \\ \left.\frac{\partial^2 C^2}{\partial\varepsilon^2}\right|_0 &= \frac{8192}{3r^4}. \end{aligned} \tag{DERIV}$$

Equation (W) shows that the Weyl term vanishes on the round spherical slice and throughout the torus family, whereas Nil has positive Weyl curvature squared at every finite positive shape. This distinction governs the shape minimization and small-radius behaviour developed below.

4.4 Comparison with the Ricci-tracefree EC curvature action

Define $\widehat{R}_{\mathrm{EC}}$ by applying only the last, Ricci-removal operation of (WP) directly to R_{EC}, with its own Ricci tensor and scalar. This is a valid projection onto the twenty-dimensional Ricci-tracefree subspace, whereas C_{EC} has ten components. The hat denotes Ricci-trace removal, not the Weyl projection. Define its squared norm as $I_{\mathrm{tf}} = (\widehat{R}_{\mathrm{EC}})_{abcd}(\widehat{R}_{\mathrm{EC}})^{abcd}$, where tf denotes zero Ricci contraction. In the standard decomposition,

$$\widehat{R}_{\mathrm{EC}} = C_{\mathrm{EC}} + R^{(2)} + R^{(3)}, \qquad I_{\mathrm{tf}} = I_{\mathrm{W}} + \|R^{(2)}\|^2 + \|R^{(3)}\|^2. \tag{TF}$$

Here $R^{(2)}$ is pair-antisymmetric and Ricci-tracefree (nine components), and $R^{(3)}$ is the four-form part (one component). Orthogonality gives the norm sum. The Ricci-tracefree EC curvature action replaces I_{W} in (A) by I_{tf}; it assigns the same quadratic coupling to all three sectors. The irreducible EC-Weyl action sets the latter two couplings to zero. Neither choice follows from the decomposition alone.

Let U_{W} and U_{tf} be the reduced potentials of the irreducible EC-Weyl and Ricci-tracefree EC curvature actions, respectively. On the specified family, (CD) gives

$$\begin{aligned} \|R^{(2)}\|^2 = \|R^{(3)}\|^2 &= \frac{8V^2\eta^2}{3r^2}, \\ I_{\mathrm{tf}} - I_{\mathrm{W}} &= \frac{16V^2\eta^2}{3r^2}, \\ U_{\mathrm{tf}} - U_{\mathrm{W}} &= -\frac{16}{3}\alpha c_M L V^2 \eta^2 r. \end{aligned} \tag{SHIFT}$$

The additional term is independent of shape and of the NY coupling. It vanishes for $V = 0$ or $\eta = 0$, and the potentials also agree at $\alpha = 0$. This contribution is determined by the selected curvature sectors and is independent of the NY coefficient correction discussed in Appendix D.

All three Ricci-tracefree EC curvature potentials follow from (S3),(TN) by replacing $9\eta^2$ by $\gamma = \eta^2(9 - 16\kappa^2\alpha V^2)$. The spherical shape difference is unchanged: at $\alpha \le 0$ the finite round minimum condition becomes $V \neq 0, \gamma < 36$, with $r^2 = (36-\gamma)/(3V^2)$; the remaining cases follow from $U_{\mathrm{tf,round}} = c_{S^3} L[V^2 r^3 + (\gamma - 36)r]/(3\kappa^2)$. At positive coupling the spherical potential is unbounded below over radius and shape. Nil has the same classification over radius and shape because (SHIFT) has no shape dependence: at nonpositive coupling $\gamma \ge 0$, its zero infimum is unattained; at positive coupling its inverse-radius term is unbounded below. For the torus, $\gamma < 0$ produces a finite radial minimum at $r^2 = -\gamma/(3V^2)$ for every shape, a degenerate minimum set in radius and shape. Otherwise the toroidal infimum is zero and unattained unless both

torsion amplitudes vanish, when every radius and shape minimizes. These comparisons concern a different action; Theorems 1–3 below describe the irreducible EC-Weyl and LC-Weyl actions.

5 Spherical shape bound and global minima

Theorem 1 (shape bound for nonpositive coupling). For $\alpha = -\beta \le 0$ and fixed L, κ, η, V, the following identities hold for every $r, x > 0$, including $\beta = 0$.

$$\begin{aligned} 3 - f(x) &= (x-1)^2(x^2+2x+3) \ge 0, \\ U_{S^3}(r,x;-\beta) - U_{S^3}(r,1;-\beta) &= \frac{c_{S^3}L}{3\kappa^2}(x-1)^2\left[12r(x^2+2x+3) + \frac{1024\kappa^2\beta x^2(x^2+x+1)^2}{r}\right]. \end{aligned} \tag{B}$$

The bracket is strictly positive, so equality occurs only at $x = 1$. The EC term and the nonnegative Weyl term both support the round minimum at each fixed radius. At $\alpha = 0$ the Weyl penalty vanishes at every shape, but the EC term still gives the strict shape difference above.

Global minimization over r, x therefore reduces to

$$\begin{aligned} b &= 9\eta^2 - 36, \qquad U_{S^3}(r,1) = \frac{c_{S^3}L}{3\kappa^2}(V^2r^3 + br), \\ r_* &= \frac{\sqrt{12-3\eta^2}}{|V|}, \qquad U_* = -\frac{2c_{S^3}LV^2r_*^3}{3\kappa^2} \quad (V \neq 0,\ \eta^2 < 4). \end{aligned} \tag{MIN}$$

When $V \neq 0$ and $\eta^2 < 4$, the radial derivative crosses zero once from negative to positive. This is the unique global radial minimum, and (B) makes the shape unique as well. The restricted Hessian in (r, ε) is positive definite:

$$\begin{aligned} H_{rr} &= \frac{2c_{S^3}LV^2r_*}{\kappa^2} > 0, \qquad H_{r\varepsilon} = 0, \\ H_{\varepsilon\varepsilon} &= \frac{c_{S^3}L}{3\kappa^2}\left(64r_* + \frac{8192\kappa^2|\alpha|}{r_*}\right) > 0. \end{aligned} \tag{H}$$

The remaining degeneracies, unattained infima, and runaway cases are classified in Section 7. Here "global" always refers to the stated fixed data and the domain $r > 0, x > 0$.

Where the minimum exists, vanishing of the Weyl term on the round slice makes its radius and value independent of nonpositive α. In contrast, the shape entry of (H) increases with $|\alpha|$. The selection of radius and the restoring curvature in the shape direction can therefore be read separately.

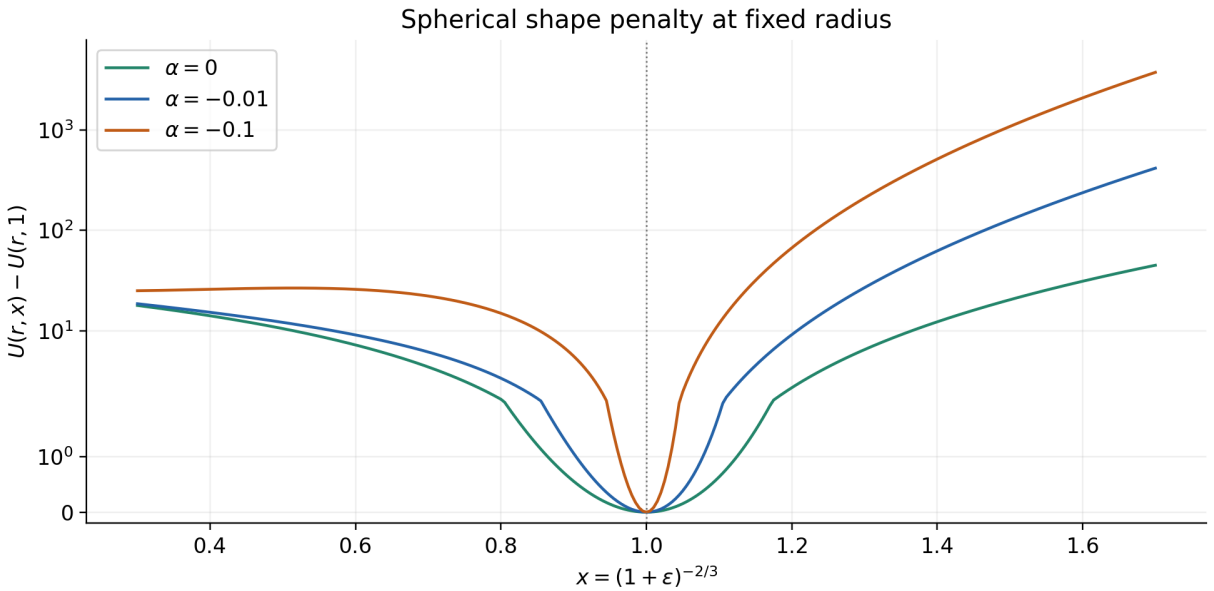


Figure 1: Spherical shape differences at $r = L = \kappa = 1$. Curves show $U(x) - U(1)$ at fixed radius; nonnegativity follows from (B).

6 Positive coupling and unboundedness

Theorem 2 (small-radius unboundedness). For $\alpha > 0$, U diverges to minus infinity as $r \to 0^+$ at any fixed $x \neq 1$ for S^3, and any fixed $x > 0$ for Nil. This holds for every fixed finite pair η, V.

$$\begin{aligned} \lim_{r\to 0^+} rU_{S^3}(r,x) &= -\frac{1024 c_{S^3} L\alpha}{3} x^2(1-x^3)^2 < 0 \quad (x \neq 1), \\ \lim_{r\to 0^+} rU_{\mathrm{Nil}^3}(r,x) &= -\frac{4 c_{\mathrm{Nil}^3} L\alpha}{3} x^8 < 0 \quad (x > 0). \end{aligned} \tag{ASY}$$

In (S3),(TN), the terms proportional to r and r^3 vanish as the radius decreases, while the inverse-radius coefficient is negative under the stated conditions. A fixed nonround shape thus suffices to establish unboundedness. On the round spherical slice the Weyl term vanishes, so its radial behaviour is distinguished from the nonround asymptotics.

Figure 2 displays the round and nonround curves and the Nil small-radius behaviour at common fixed parameters. Equation (ASY) supplies the full-domain statement corresponding to these finite samples.

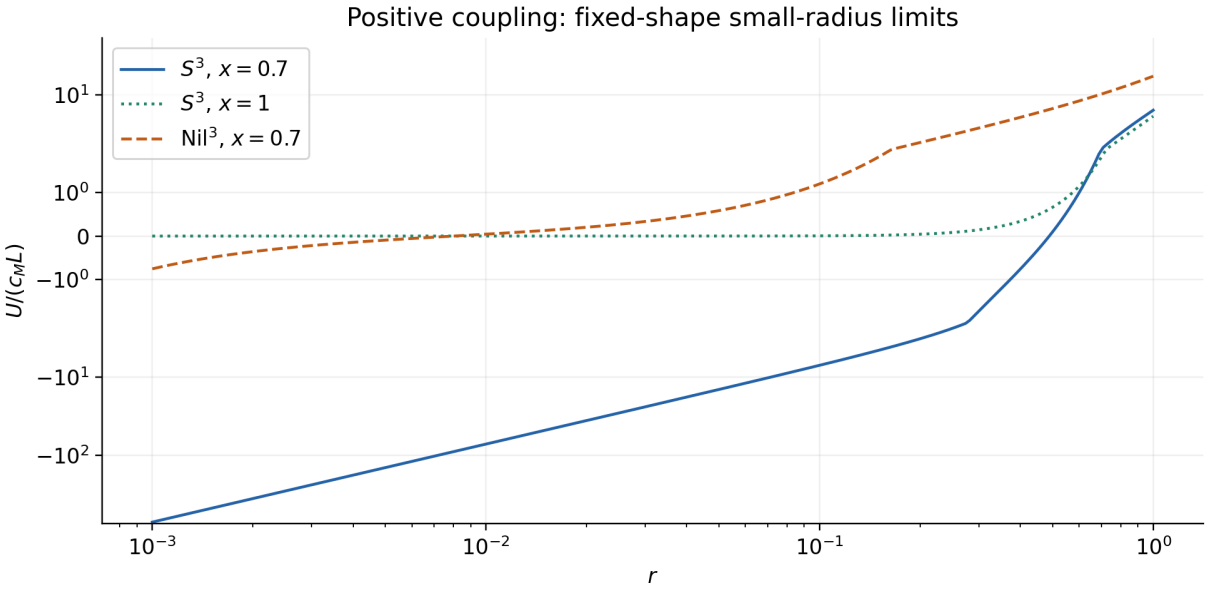


Figure 2: Small-radius behaviour at $V_\tau = 4$, $\eta_\tau = -2$, $L = \kappa = 1$, $\alpha = 0.01$, $x = 0.7$. Finite samples illustrate (ASY).

7 Parameter classification and representative slices

Theorem 3 (complete spherical classification at fixed torsion parameters). For $\alpha > 0$ all parameter choices are unbounded below. For $\alpha \leq 0$, (MIN) gives the table. An unattained zero infimum is approached as $r \to 0$; the unbounded case with $V = 0, \eta^2 < 4$ follows along the round ray $r \to \infty$.

Fixed data ($\alpha \leq 0$)	Infimum and attainment
$V \neq 0,\ \eta^2 < 4$	Unique negative global minimum (MIN)
$V \neq 0,\ \eta^2 \geq 4$	Zero infimum, unattained
$V = 0,\ \eta^2 < 4$	Unbounded below
$V = 0,\ \eta^2 = 4$	Zero minimum on the entire round ray
$V = 0,\ \eta^2 > 4$	Zero infimum, unattained

At $V = 0, \eta^2 = 4$, the potential is identically zero on the round ray; there is no isolated radial minimum. The sign of α alone therefore determines neither existence of a minimum nor boundedness for all parameters on its nonpositive side.

The competition between the cubic term controlled by V and the linear term controlled by η determines the radial extrema. When V vanishes the cubic term is absent, changing the $\eta^2 < 4$ case from a finite-radius minimum to a large-radius runaway. At $\eta^2 = 4$ the linear term vanishes: nonzero V gives monotonic increase, while $V = 0$ leaves the entire round ray as the minimum set.

The next table fixes $V_\tau = 4$, $L = \kappa = 1$ and compares amplitudes inside, on, and outside the threshold. The round values are common to all nonpositive α, and reversing the sign of η leaves the potential unchanged.

Code η_τ (geometric η)	Round slice ($V_\tau = 4$, $L = \kappa = 1$)
-3 (3)	No positive stationary radius
-2 (2)	No positive stationary radius
0 (0)	$r_* = \sqrt{3}/2,\ U_* = -\sqrt{3}\pi^2$
2 (-2)	No positive stationary radius

The $\eta_\tau = 0$ slice realizes the finite-radius minimum in (MIN); $\eta_\tau = \pm 2$ lies on the threshold, and $\eta_\tau = -3$ outside it. Appendix D displays these curves together with their relation to the preceding version's reference expression.

8 Comparison with the torus and Nil families

For T^3 both LC-Weyl and complete NY vanish. U is independent of α, θ, and shape. The shape independence relies on the shape-independent axial coefficient in (T), not on flatness alone. If either torsion amplitude is nonzero, it increases strictly with radius and has unattained infimum zero. If both vanish it is zero throughout, selecting neither a radius nor a shape.

For Nil and $\alpha \leq 0$,

$$\frac{\partial U_{\mathrm{Nil}^3}}{\partial x} = \frac{c_{\mathrm{Nil}^3} L}{3\kappa^2}\left(3x^3 r + \frac{32\kappa^2\beta x^7}{r}\right) > 0 \qquad (\alpha = -\beta \leq 0). \tag{NIL}$$

There is no minimum at finite shape in the two-variable problem. The positive potential approaches infimum zero along $r = \ell t, x = t$, $t \to 0^+$, with fixed positive length ℓ. A separate radial minimum exists for negative α when shape is fixed:

$$\begin{aligned} A = V^2, \quad B = 9\eta^2 + \frac{3}{4}x^4, \quad D = 4\kappa^2\beta x^8, \\ r_{\mathrm{slice}}^2 = \frac{2D}{B + \sqrt{B^2 + 12AD}} \qquad (\beta > 0). \end{aligned} \tag{SLICE}$$

This expression includes $A = 0$ but requires $\beta > 0$. Releasing shape always lowers the potential in the direction specified by (NIL), so this radial minimum is not a two-variable minimum. Positive α is covered by Theorem 2. At the common API amplitudes $V_\tau = 4, \eta_\tau = -2$ and nonpositive α, all three infima are zero and none is attained at an interior point. These inputs illustrate comparison among three unattained infima. For $V \neq 0, \eta^2 < 4$, the negative attained spherical minimum can be compared with the two zero infima, but only for these specified quotients and fixed-amplitude paths, not as a universal topology-selection law. Vanishing local Nil curvature in a shape limit does not turn the fixed compact quotient into a torus.

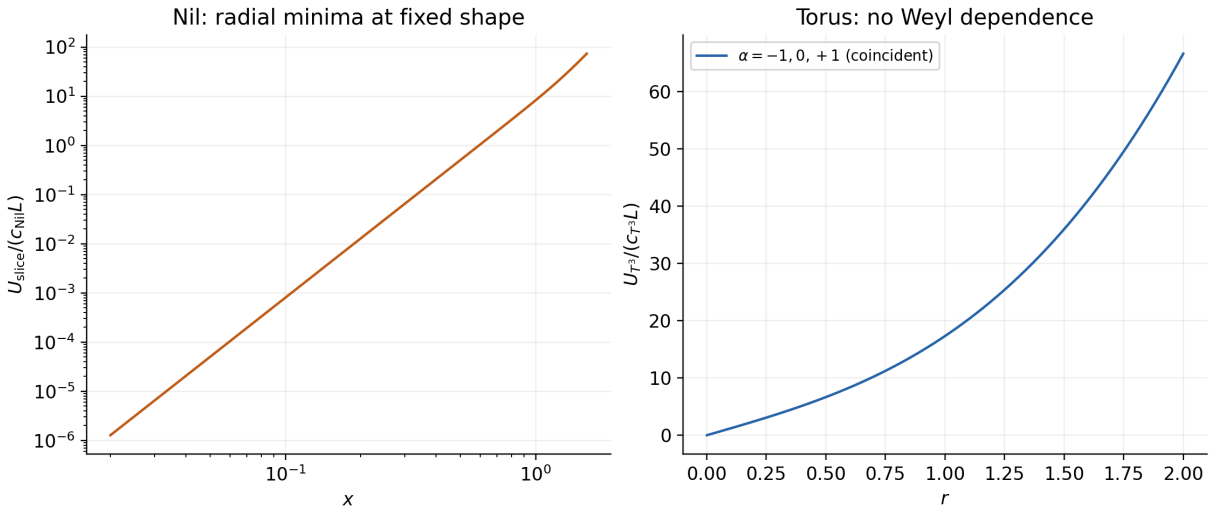


Figure 3: Left: values of Nil radial slice minima approaching the unattained infimum ($\alpha = -1$). Right: toroidal alpha independence. $V_\tau = 4$, $\eta_\tau = -2$, $L = \kappa = 1$.

9 Variational scope and physical questions

We have held L, η, V fixed. If these are varied, complete NY gives additional necessary conditions:

$$\frac{\partial U_M}{\partial V} = \frac{2c_M L V r^3}{3\kappa^2}, \qquad \frac{\partial U_M}{\partial \eta} = \frac{6c_M L \eta r}{\kappa^2}, \qquad \frac{\partial U_M}{\partial L} = \frac{U_M}{L}. \tag{VAR}$$

For example, the negative spherical minimum with $V \neq 0, \eta^2 < 4$ has nonzero derivatives with respect to V and L. It is not even stationary after those fixed data are released within the reduced family. A positive restricted Hessian is not full EC stability. Conversely, stationarity in all reduced parameters alone would not establish all unreduced field equations.

This distinction locates the shape bound and Hessian as measures of the geometric response within the specified variation space. A full classical solution must satisfy the field equations and constraints, including those for the independent connection.

Dynamical stability requires a treatment of time-dependent perturbations, kinetic terms, and constraints. The Hamiltonian instability considered in Ostrogradsky's theorem[5] and boundedness of the present static Euclidean potential are different criteria. Theorem 2 is accordingly a statement about the restricted action at small radius.

Likewise, the self-duality constraint within the spatial two-form subspace in Proposition 1 and the existence of general quantum transition paths are different questions. Analysis of transitions with nonhomogeneous fields and boundary conditions extends beyond the static homogeneous family. Relating comparisons of action values to physical selection also requires common constraints and observables across the sectors being compared.

The derivative identities (VAR) concern the irreducible EC-Weyl and LC-Weyl actions on this family. Equality of restricted potentials does not imply equality of full-field equations or kinetic operators. The Ricci-tracefree EC curvature action is distinct; its extra term (SHIFT) must also be differentiated when varying fixed torsion data.

10 Conclusions

We have analysed static homogeneous reductions of the EC plus complete NY plus irreducible EC-Weyl-squared action in radius and volume-preserving shape. The complete NY density and form-index Pontryagin density vanish for the prescribed connection, and common conventions give explicit curvature invariants and potentials for the sphere, torus, and Nil families.

The tensor identity $C_{\mathrm{EC}} = C_{\mathrm{LC}}$ holds for every positive shape in the prescribed torsion family. The two Weyl actions share the same reduced potential and classification. The Ricci-tracefree EC curvature action includes two additional curvature sectors and has the distinct potential shift (SHIFT); its effects are not attributed to the irreducible Weyl sector.

The central spherical shape factorization reduces global minimization at nonpositive Weyl coupling to the round slice. For $V \neq 0$ and $\eta^2 < 4$ there is a unique negative finite-radius minimum. Its radius and value are independent of the Weyl coupling, while the shape entry of the Hessian increases with the coupling's absolute value. The remaining parameter regions, including degenerate cases, separate unattained infima, a minimum set on the whole round ray, and a large-radius runaway.

Positive Weyl coupling gives small-radius unboundedness for nonround spheres and Nil. The torus potential is independent of shape and Weyl coupling, while monotonicity in Nil shape excludes a finite-shape two-variable minimum on the nonpositive side. Distinguishing fixed-shape radial minima from full shape minimization enables an analytic comparison of the extrema across the geometries.

These results identify how the curvature-squared term acts on radius and shape at fixed torsion parameters and circle circumference. Extensions to full-field vacua, time-dependent stability, and quantum transitions require the additional analysis of fields, constraints, and observables described in Section 9.

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For this revision, Codex (OpenAI; accessed 2026-10) also assisted with computational and plotting code, bilingual presentation, and symbolic and numerical checks. Numerical values and plots were obtained by deterministic evaluations of the stated formulas and checked against the accompanying implementation; they are not synthetic images produced by a generative image model. The author takes full responsibility for the content and for any remaining errors.

This work was developed within the informal collaborative project **DPPU** (**D**onut-like topology, **P**lanck-scale compactness, **P**recession dynamics, **U**niverse).

A Local connection and the input dictionary

Solving metric compatibility and zero torsion gives $\Gamma^{\mathrm{LC}}_{abc} = (C_{abc} + C_{cba} - C_{bac})/2$. The last connection index is the derivative direction. Reconstruct geometric torsion as $T_{abc} = C_{abc} + \Gamma_{acb} - \Gamma_{abc}$. Substitution of (D) gives the same connection in the manuscript and in the code.

For constant coefficients in this noncoordinate frame,

$$R^a{}_{bcd} = \Gamma^a{}_{ec}\Gamma^e{}_{bd} - \Gamma^a{}_{ed}\Gamma^e{}_{bc} + \Gamma^a{}_{be}C^e{}_{cd}. \tag{CURV}$$

The structure-constant term is essential. Parameter domains transform with their signs: code $V_\tau > 0$ means geometric $V < 0$, and $\eta_\tau \in [a, b]$ maps to $\eta \in [-b, -a]$. The scale, length, and couplings do not change. Invariance of scalar results under simultaneous sign reversal does not permit discarding the signs of torsion or the three-form.

The spherical scalar $24/r^2$ fixes physical radius $r/2$ and its global volume. The product of the three shape factors is one. Nil uses the domain and gluing in Section 2.

B Reproduction and classification outputs

The public entry point is `script/README.md`. The render command of `reproduce_figures.py` draws Figures 1–4 from the supplied data; recompute uses the public engine to write six figure/table CSV files, parameters, and images to a new directory. Only the Appendix D comparison curve uses the preceding NY expression with the current spherical volume. The three geometry runners and two symbolic supplements reproduce the potentials, LC-Weyl expression, and EC curvature block structure. The conventions guide specifies indices, signs, global geometry and shape; the implementation guide explains symbolic computation and APIs, and the classification guide defines scan/viewer outputs. The public library supports these current reproduction paths; the scientific arguments are given in the paper and appendices.

The scan and viewer share a classifier. For complete NY, it separately reports the Sections 7–8 classification on all $r, x > 0$ and the radial classification on all $r > 0$ at fixed shape. Torsion amplitudes, L and κ are fixed; the CLI/viewer use $L = \kappa = 1$. Finite minima, unattained infima, degenerate minimum sets, and unbounded potentials are distinguished. A round spherical slice can have a minimum at positive α while the domain with both radius and shape varying is unbounded. A Nil minimum at fixed shape is not a minimum over radius and shape. Actions using $\mathcal{T}$ or $\mathcal{R}$ alone are outside this classification, but their potential curves remain available for comparison.

For fixed shape, (S3),(TN) give

$$\begin{aligned}
U(r) &= p(Ar^3 + Br + D/r), \quad p = \frac{c_M L}{3\kappa^2} > 0, \quad A = V^2, \\
(B, D)_{S^3} &= (9\eta^2 - 12f(x), -1024\kappa^2\alpha x^2(1 - x^3)^2), \\
(B, D)_{T^3} &= (9\eta^2, 0), \\
(B, D)_{\mathrm{Nil}^3} &= (9\eta^2 + 3x^4/4, -4\kappa^2\alpha x^8), \\
U'(r) = 0 &\iff 3Ay^2 + By - D = 0, \quad y = r^2 > 0.
\end{aligned} \tag{RAD}$$

The second derivative at positive roots and the limits at zero and infinity determine the infimum and attainment. If $D < 0$, the small-radius potential is unbounded below even when a local

radial minimum exists. If $A = B = D = 0$, all radii are degenerate minima. Plot windows do not discard roots; numerical grids only visualize classifications.

Inputs are treated as their represented floating-point values, without rounding to two decimal places. The boundaries $V = 0$, $|\eta| = 2$ and $\alpha = 0$ are not identified with neighbouring values. Fixed-shape coefficients and roots use cancellation-resistant expressions and high-precision arithmetic, with unresolved boundaries and unrepresentable numeric values explicitly flagged. The viewer displays current engine curves alongside both domain classifications, infima and radii. These describe restricted potentials, not full-field vacua or Lorentzian ghosts. The classification guide in the attached script directory specifies the CSV fields and their classification domains.

The public selectors are `EC_IRR` (default) for the irreducible EC-Weyl action, `LC` for the LC-Weyl action, and `EC_TRACEFREE` for the Ricci-tracefree EC curvature action. The six CSVs, parameters and four figures retain their LC provenance; recomputation explicitly selects LC. By (ID) they also apply to the irreducible EC-Weyl action on this family. The Ricci-tracefree EC curvature action replaces $9\eta^2$ by γ in (RAD), with the classification in Section 4.4. At its toroidal minimum every shape minimizes: the CSV field `global_r` records the common radius and `global_x` is empty. Threshold signs use exact rational arithmetic on the supplied binary floats before numerical evaluation.

The torsion selectors allow the axial component alone (`AX`, $V = 0$), the vector-trace component alone (`VT`, $\eta = 0$), or both components (`MX`). Zero amplitudes are allowed in all modes. Independently, `FULL` uses $\mathcal{T} - \mathcal{R}$ from (NYD), while `TT` and `REE` use $+\mathcal{T}$ and $+\mathcal{R}$, respectively, multiplied by the real coupling θ. The plus sign denotes the component convention, not a restriction on θ. In particular, the standalone `REE` choice does not include the minus sign in the complete NY form.

C Proof details and normalizations

Equation (R) follows by solving the Cartan equations for general n_i and using (CURV). The torsion contribution is $-6\eta^2/r^2 - 2V^2/3$; no term linear in both n_i and η survives.

Equation (W) follows from LC curvature and Ricci norms. Identity (B) uses $1 - x^3 = -(x-1)(x^2 + x + 1)$ and is valid for all positive x, independently of finite numerical sampling. The derivative of the round polynomial is $3V^2r^2 + b$, which yields the classification including its degenerate cases.

The Pontryagin coefficient is defined by exterior products, retaining its factor of two relative to the all-component sum. Zero examples alone cannot check a normalization. For the algebraic control $R_{01} = e^0 \wedge e^1 + 2e^2 \wedge e^3$, $R_{10} = -R_{01}$, with all other entries zero, $\mathcal{P} = 8$ and $P_{\text{all}} = 16$. This is an array-normalization control, not a background solution.

D Revision note and reference curves

This version uses coefficient $1/2$ in the curvature-side component expression of complete NY and specifies global volumes and the torsion-input dictionary. The curvature identities and Weyl analysis are organized into the fixed-parameter classification of Theorems 1–3. The comparison below isolates the effects on the reference potential. Section 9 states the variational scope and limits of physical interpretation.

The preceding reference inputs were $V_\tau = 4$, $\eta_\tau = -2$, $\theta = L = \kappa = 1$, corresponding to geometric $V = -4$, $\eta = 2$. Equation (REF) and the left panel of Figure 4 compare the old NY expression with complete NY at the same present volume.

$$U_{\text{old,scaled}}(r) = \frac{4\pi^2}{3} r^2 (r - 3), \qquad U_{\text{corrected}}(r) = \frac{4\pi^2}{3} r^3. \tag{REF}$$

The complete-NY curve increases strictly at positive radius, so the old expression's minimum at $r = 2$ is absent. Published action values also used a spherical volume eight times the present value; the component correction and the volume change must therefore be distinguished. The preceding toroidal volume was also 2π times the present one. The right panel shows the four slices in Section 7. Its $\eta_\tau = 0$ minimum realizes a different fixed-parameter condition, rather than a replacement physical vacuum for the preceding reference point.

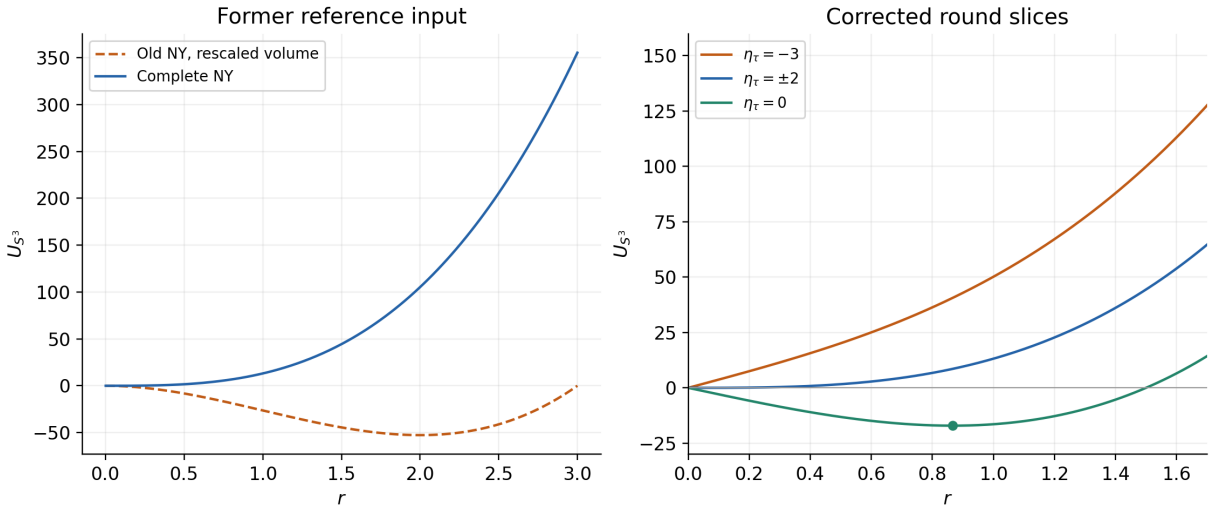


Figure 4: Left: old-NY and complete-NY curves at the same current volume and preceding reference inputs. Right: round slices for the four inputs in Section 7.

Figures 1–4 use the current formulas and the explicitly identified comparison expression. They should not be matched by number alone to the twelve figures of the preceding version.

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